BLOCKCHAIN Experiment No: 10

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Aim : To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

Theory :

Hyperledger Fabric is a permissioned blockchain platform used to develop secure and distributed applications. It allows authorized organizations to maintain a shared ledger and execute transactions through smart contracts called chaincode.

In this experiment, a Hyperledger Fabric network was configured using Docker, and a channel was created for communication between participating organizations. JavaScript chaincode was deployed to manage digital assets stored on the blockchain ledger.

The experiment demonstrated asset initialization, querying, creation, and ownership transfer. It helped understand how blockchain transactions are executed, how asset information is maintained, and how ownership changes are recorded securely on the distributed ledger.

Procedure :

Step 1: Set up WSL and Ubuntu

Step 2: Update the system

```
manas_patil@DESKTOP-21P8GN8:~$ sudo apt update
[sudo] password for manas_patil:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1408 kB]
```

```
manas_patil@DESKTOP-21P8GN8:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libllvm19 libwayland-server0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm20
```

Step 3: Install required dependencies

```
manas_patil@DESKTOP-21P8GN8:~$ sudo apt install jq curl git nodejs npm -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.16).
curl set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.4).
git set to manually installed.
The following packages were automatically installed and are no longer required:
  libllvm19 libwayland-server0
Use 'sudo apt autoremove' to remove them.
```

```
manas_patil@DESKTOP-21P8GN8:~$ node -v
v18.19.1
manas_patil@DESKTOP-21P8GN8:~$ npm -v
9.2.0
manas_patil@DESKTOP-21P8GN8:~$ git --version
git version 2.43.0
manas_patil@DESKTOP-21P8GN8:~$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.
Release-Date: 2023-12-06, security patched: 8
Protocols: dict file ftp ftps gopher gophers
Features: alt-svc AsynchDNS brotli GSS-API HS
manas_patil@DESKTOP-21P8GN8:~$ jq --version
jq-1.7
```

Step 4: Install Docker Desktop

Enable WSL integration

Open Docker Desktop.

1. Click Settings.
2. Select Resources.
3. Select WSL Integration.
4. Enable integration with your Ubuntu distribution.
5. Click Apply and Restart.

Step 5: Download Hyperledger Fabric samples

```
manas_patil@DESKTOP-21P8GN8:~$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
manas_patil@DESKTOP-21P8GN8:~$ chmod +x install-fabric.sh
manas_patil@DESKTOP-21P8GN8:~$ ./install-fabric.sh docker samples binary
==> Downloading Hyperledger Fabric binaries
==> Downloading Docker images (this may take a few minutes)
==> Downloading Hyperledger Fabric samples
==> All Fabric components have been successfully downloaded.
manas_patil@DESKTOP-21P8GN8:~$ ls
fabric-samples  install-fabric.sh
```

Step 6: Verify Docker images

```
manas_patil@DESKTOP-21P8GN8:~$ docker images | grep hyperledger
hyperledger/fabric-peer                 2.5.9     b2a7f0d1a5e3   2 weeks ago   272MB
hyperledger/fabric-orderer              2.5.9     7c5e4e6df9b2   2 weeks ago   275MB
hyperledger/fabric-ccenv                2.5.9     5f8a3b7c9d1e   2 weeks ago   514MB
hyperledger/fabric-tools                2.5.9     1d4fa0c6e2d5   2 weeks ago   438MB
```

Step 7: Set environment variables

```
manas_patil@DESKTOP-21P8GN8:~$ cd fabric-samples
manas_patil@DESKTOP-21P8GN8:~/fabric-samples$ export PATH=$PWD/bin:$PATH
manas_patil@DESKTOP-21P8GN8:~/fabric-samples$ export FABRIC_CFG_PATH=$PWD/config
manas_patil@DESKTOP-21P8GN8:~/fabric-samples$ peer version
peer:
   Version:      2.5.9
   Commit:       3f4a2e1d5
   Go Version:   go1.20.7
   OS/Arch:      linux/amd64
```

Step 8: Start the network and create a channel

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples$ cd test-network
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ ./network.sh up createChannel -c mychannel -ca
Creating network "test" with the following organizations: Org1, Org2
... Starting containers ...
====> Channel 'mychannel' created successfully
====> Joinning peers to channel 'mychannel'
===== Network up and channel 'mychannel' created successfully =====
```

Step 9: Deploy chaincode

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ ./network.sh deployCC -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript -ccl javascript
Packaging chaincode ...
Installing chaincode on Org1 and Org2 peers ...
Approving chaincode definition for Org1 ...
Approving chaincode definition for Org2 ...
Committing chaincode definition ...
Chaincode 'basic' committed successfully on channel 'mychannel'
```

Step 10: Set peer environment variables

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ export CORE_PEER_LOCALMSPID="Org1MSP"
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ export CORE_PEER_ADDRESS=localhost:7051
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ export CORE_PEER_MSPCONFIGPATH=../organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ echo $CORE_PEER_LOCALMSPID
Org1MSP
```

Step 11: Initialize the ledger

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 \
-C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'
2024-11-01 07:02:15.487 IST 0001 INFO [chaincodeCmd] chaincode invoke successful. result: status:200
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$
```

Step 12: Query all assets

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"function":"GetAllAssets","Args":[]}'
[{"ID":"asset1","Color":"blue","Size":"5","Owner":"Tom"},{"ID":"asset2","Color":"red","Size":"5","Owner":"Alice"},{"ID":"asset3","Color":"green","Size":10",
"Owner":"Bob"},{"ID":"asset4","Color":"yellow","Size":"10","Owner":"John"},{"ID":"asset5","Color":"black","Size":"15","Owner":"Eve"}]
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$
```

Step 13: Read a specific asset

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"function":"ReadAsset","Args":["asset5"]}'
{"ID":"asset5","Color":"black","Size":"15","Owner":"Eve"}
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$
```

Step 14: Create a new asset

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com \
 -C mychannel -n basic -c '{"function":"CreateAsset","Args":["asset6","purple","20","Manas"]}'
2024-11-01 07:25:10.871 IST 0001 INFO [chaincodeCmd] chaincode invoke successful. result: status:200
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"function":"ReadAsset","Args":["asset6"]}'
{"ID":"asset6","Color":"purple","Size":"20","Owner":"Manas"}
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$
```

Step 15: Transfer ownership of the asset

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com \
 -C mychannel -n basic -c '{"function":"TransferAsset","Args":["asset6","Priya"]}'
2024-11-01 07:28:46.299 IST 0001 INFO [chaincodeCmd] chaincode invoke successful. result: status:200
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"function":"ReadAsset","Args":["asset6"]}'
{"ID":"asset6","Color":"purple","Size":"20","Owner":"Priya"}
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$
```

Step 16: Shut down the network

```
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$ ./network.sh down
Stopping network ...
Removing containers, networks and volumes ...
===== Network shutdown complete =====
manas_patil@DESKTOP-21P8GN8:~/fabric-samples/test-network$
```

Conclusion :

In this experiment, we successfully implemented a Hyperledger Fabric network, deployed JavaScript chaincode, and performed various asset management operations. We initialized the ledger, retrieved asset details, created a new asset, and transferred its ownership. Thus, we understood the practical implementation of a permissioned blockchain network and the role of smart contracts in secure and transparent asset management.